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Surface energy and the contact of elastic solids

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This paper discusses the influence of surface energy on the contact between elastic solids. Equations are derived for its effect upon the contact size and the force of adhesion between two lightly loaded spherical solid surfaces. The theory is supported by experiments carried out on the contact of rubber and gelatine spheres.

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Introduction

There is much evidence to suggest that attractive forces act between solids close together (see, for example, Adamson 1967) and such forces explain qualitatively why a mechanical load is required to separate two solid bodies placed in intimate contact. Extensive measurements have been made of the range of action of these surface forces (for a review see Tabor & Winterton 1969). Measurements have also been made of the total mechanical load required to separate adhering bodies but attempts to relate these measurements to surface forces acting between body interfaces have not been very successful. Mechanical adhesive measurements include studies of glues, paints and vacuum evaporated coatings (see, for example, Alner 1969). Associated with every surface is a surface energy resulting from the action of surface forces. Measurements of this have been mainly confined to liquids though some estimates have been made for mica, quartz and glass (Bradley 1932; Bailey 1957, 1961). The surface energy and strength of adhesion between elastic bodies are clearly related through the action of surface forces, but in a way that is not always obvious.

In order to separate bodies in intimate contact mechanical work must be expended to overcome the adhesive forces. This work goes to create 'new' surface. The energy required to create unit area of new surface can be defined as the free surface energy of the solid. Although this is a straightforward definition it is difficult to find direct experimental evidence in support of it (Bikerman 1965). The behaviour of solids differs markedly from that of liquids. The spreading or contracting of one liquid surface over another or over a solid surface to reach an equilibrium state is dominated by the minimization of surface energy. For example, when a liquid drop is brought into contact with a solid surface its final contact size at equilibrium may be predicted from surface energy considerations (Good 1952). On the other hand when a contact is formed between two smooth solid surfaces the

equilibrium largely depends upon the distribution of elastic forces in the contacting bodies.

However, under conditions of light loading between elastic solids surface forces can make a significant contribution to the contact equilibrium and this paper describes a new theoretical argument for these conditions. Based upon simple observations of the contact between elastic solids an approximate theory is developed and out of this an exact analysis derived. The analysis is fully supported by simple experiments and suggests a possible re-interpretation of some of the published measurements on the force of adhesion between solid bodies.

Observations on solid contact

The contact between two smooth elastic bodies was investigated by Hertz (1896) who demonstrated that both the size and shape of the zone of contact followed from the elastic deformation of the bodies. For two spheres of radius R_1 and R_2 pressed together under a load P_0 (see figure 1 *a*) the radius a_0 of the circle of contact is given by

$$a_0^3 = \tfrac{3}{4}\pi(k_1+k_2)\frac{R_1R_2}{R_1+R_2}P_0, \tag{1}$$

where k_1 and k_2 are the elastic constants of the material of each sphere; that is

$$k_1 = \frac{1-\nu_1^2}{\pi E_1} \quad \text{and} \quad k_2 = \frac{1-\nu_2^2}{\pi E_2}$$

where ν is the Poisson ratio and E the Young modulus of each material. Resulting from local compression near to the contact region, distant points in the two spheres approach each other by a distance δ given by

$$\delta^3 = \tfrac{9}{16}\pi^2(k_1+k_2)^2\frac{R_1+R_2}{R_1R_2}P_0^2. \tag{2}$$

Hertz used an optical microscope to measure the contact between glass spheres and so verified his theory experimentally.

Recently, however, some experimental contradictions to the Hertz theory have been reported. Roberts (1968*a*) using smooth rubber spheres and Kendall (1969) using glass spheres noted that at low loads contact areas between these bodies were considerably larger than those predicted by Hertz and tended towards a constant finite value as the load was reduced to zero. Strong adhesion was observed between them if the surfaces were clean and dry. At high loads the results closely fitted the Hertz theory.

These observations strongly suggested that attractive surface forces were operating between the solids and although these 'additional' contact forces were of little significance at high loads they became increasingly important as the load was reduced towards zero.

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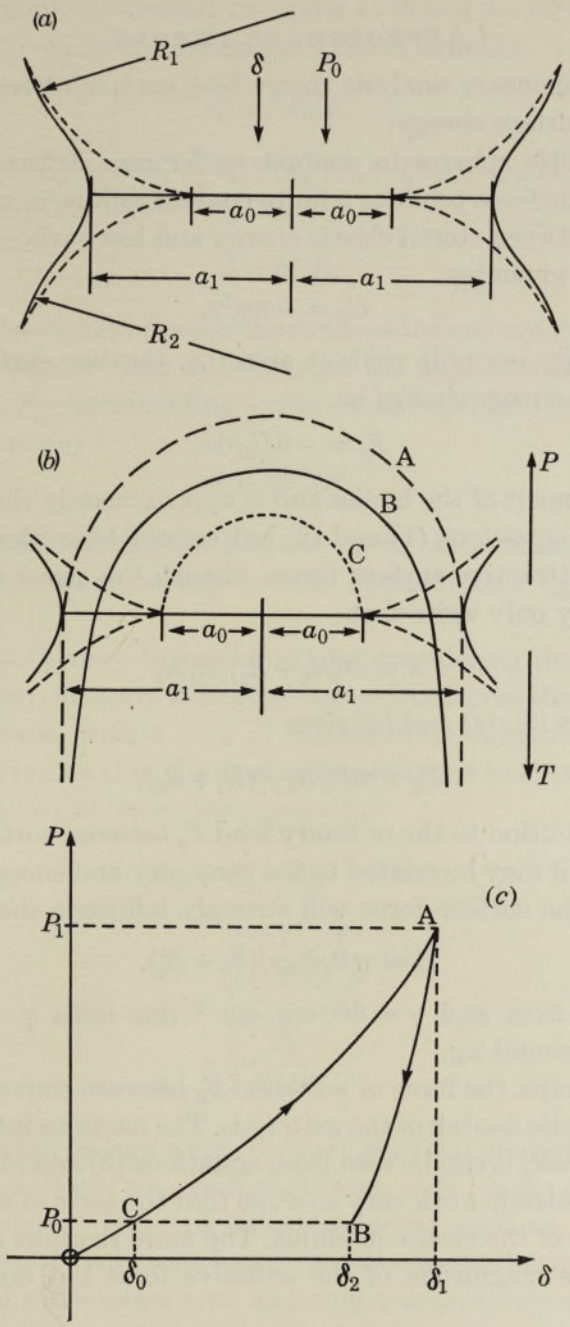


FIGURE 1. The contact between two elastic solids both in the presence (contact radius a_1) and absence (contact radius a_0) of surface forces. (*a*) shows the contact between two convex bodies of radii R_1 and R_2 under a normal load of P_0; δ is the elastic displacement. (*b*) indicates the distribution of stress in the contacting spherical surfaces. When surfaces are maintained in contact over an enlarged area by surface forces the stresses between the surfaces are tensile (T) at the edge of the contact and only remain compressive (P) in the centre. Distribution A is the Hertz stress with $a = a_1$ and $P = P_1$; distribution B the actual stress (Johnson 1958) with $a = a_1$ and $P = P_0$ and distribution C the Hertz stress with $a = a_0$ and $P = P_0$. (*c*) represents the load-displacement relation for the contacting surfaces.

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Approximate theory

The following elementary analysis shows how surface attraction may be interpreted in terms of surface energy.

Consider two elastic spheres in contact under zero external load. Attractive forces between the surfaces produce a finite contact radius, a, a balance eventually being established between stored elastic energy and lost surface energy. The loss in surface energy U_S is given by

$$U_S = -\pi a^2 \gamma, \tag{3}$$

where γ is the energy per unit contact area (i.e. the two surfaces). The force F_S associated with this energy change is

$$F_S = -\mathrm{d}U_S/\mathrm{d}x, \tag{4}$$

where x is the movement of the bodies and is approximately the same as δ which is given by the Hertz equations (1) and (2) but cannot be worked out exactly from these because the attractive surface forces disturb the stress distributions in the bodies. Thus we may only write that

$$x \approx a^2(R_1+R_2)/R_1R_2 \tag{5}$$

combining equations (3), (4) and (5) gives

$$F_S \approx \pi R_1 R_2 \gamma/(R_1+R_2). \tag{6}$$

This force acts in addition to the ordinary load P_0 between surfaces and the simple analysis shows that it may be related to the geometry and energy of the contacting surfaces. Further, the surface force will strongly influence the contact size when

$$P_0 \approx \pi R_1 R_2 \gamma/(R_1+R_2). \tag{7}$$

Suppose $R_1 = R_2 = 2\,\mathrm{cm}$ and $\gamma = 600\,\mathrm{erg\ cm^{-2}}$ (for mica $\gamma \approx 300\,\mathrm{erg\ cm^{-2}}$ per surface) then P_0 is around 2 g.

Surprisingly, perhaps, the force of adhesion F_S between convex surfaces does not depend upon the elastic moduli of the materials. The modulus influences the contact radius, a, but, however, it can be seen from equations (3) and (5) that both the surface energy and the elastic work vary as a^2, so that the force of adhesion is independent of a and hence of the elastic modulus. The more rigorous analysis which now follows provides the magnitude of the adhesive force but does not change this conclusion.

Exact analysis

The above theory can only be approximate since the Hertz equations are not valid when surface forces act at a contact. A rigorous determination of the contact equilibrium between elastic spheres involves computation of the total energy U_T in the system as a function of contact radius a. Equilibrium will then obtain when

$$\mathrm{d}U_T/\mathrm{d}a = 0. \tag{8}$$

Consider the situation illustrated in figure 1. When no surface forces act, the contact radius a_0 is given by the generalized Hertz equation

$$a_0^3 = RP_0/K, \tag{9}$$

where $R = R_1R_2/(R_1+R_2)$ and $K = 4/3\pi(k_1+k_2)$. The movement δ of the applied load is given by (point C in figure 1*c*)

$$\delta = a^2/R. \tag{10}$$

Now if attractive forces act between the surfaces the contact radius in equilibrium will be a_1, which is greater than a_0. Although the applied load remains at P_0, an apparent Hertz load P_1 corresponding to the contact radius a_1 may be defined such that (point A in figure 1*c*)

$$a_1^3 = RP_1/K. \tag{11}$$

The total energy U_T of this system is made up of three terms, the stored elastic energy U_E, the mechanical energy in the applied load U_M and the surface energy U_S.

The elastic energy U_E may be calculated by considering the idealised load displacement curve shown in figure 1. Neglecting surface forces the system is loaded to give a contact radius a_1 with a load P_1 (condition A) which requires energy U_1. Keeping the contact radius at a_1 the load is then reduced to P_0 to give the final state of the system (condition B), releasing energy U_2

$$U_E = U_1 - U_2,$$

$$U_1 = \int_0^{P_1} \frac{2}{3} \frac{P^{\frac{2}{3}}}{K^{\frac{2}{3}}R^{\frac{1}{3}}} . \mathrm{d}P$$

$$= \frac{2}{5} \frac{P_1^{\frac{5}{3}}}{K^{\frac{2}{3}}R^{\frac{1}{3}}}.$$

We now introduce the feature which distinguishes this analysis from the approximate theory. In that theory the contact stresses, even in the enlarged area, were assumed to be Hertzian. However when spherical surfaces are maintained in contact over an enlarged area by surface forces the stresses between the surfaces are *tensile* at the edge of the contact area and only remain compressive in the centre. The exact distribution of stress has been calculated by Johnson (1958) and is shown by curve B in figure 1*b*. The deformed profile of each sphere immediately outside the contact area is also changed as shown in figure 1*a*, from one which meets the interface tangentially (without adhesive forces (broken line)) to one which meets the interface perpendicularly with adhesive forces (full line). This change in profile was indeed observed by optical interferometry. The load-displacement relation is then given by

$$\delta = \tfrac{2}{3}P/Ka_1.$$

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Therefore
$$U_2 = \int_{P_0}^{P_1} \frac{2}{3}\frac{P}{Ka_1}\,\mathrm{d}P$$
$$= \frac{1}{3K^{\frac{2}{3}}R^{\frac{1}{3}}}\left[\frac{P_1^2 - P_0^2}{P_1^{\frac{1}{3}}}\right].$$

Thus
$$U_{\mathrm{E}} = \frac{2}{5}\frac{P_1^{\frac{5}{3}}}{K^{\frac{2}{3}}R^{\frac{1}{3}}} - \frac{1}{3}\frac{(P_1^2 - P_0^2)}{K^{\frac{2}{3}}R^{\frac{1}{3}}P_1^{\frac{1}{3}}}$$
$$= \frac{1}{K^{\frac{2}{3}}R^{\frac{1}{3}}}[\tfrac{1}{15}P_1^{\frac{5}{3}} + \tfrac{1}{3}P_0^2P_1^{-\frac{1}{3}}]. \tag{12}$$

The mechanical potential energy U_{M} of the applied load P_0 is
$$U_{\mathrm{M}} = -P_0\delta_2$$
$$= -P_0\left[\delta_1 - \frac{2}{3}\frac{(P_1 - P_0)}{Ka_1}\right]$$
$$= -P_0\left[\frac{P_1^{\frac{2}{3}}}{K^{\frac{2}{3}}R^{\frac{1}{3}}} - \frac{2}{3}\left(\frac{K}{RP_1}\right)^{\frac{1}{3}}\frac{(P_1 - P_0)}{K}\right]$$
$$= \frac{-P_0}{K^{\frac{2}{3}}R^{\frac{1}{3}}}[\tfrac{1}{3}P_1^{\frac{2}{3}} + \tfrac{2}{3}P_0P_1^{-\frac{1}{3}}]. \tag{13}$$

The surface energy U_{S} is given by
$$U_{\mathrm{S}} = -\gamma\pi a_1^2$$
$$= -\gamma\pi(RP_1/K)^{\frac{2}{3}}, \tag{14}$$

where γ is the energy of adhesion of *both* surfaces.

The total energy U_{T} is
$$U_{\mathrm{T}} = U_{\mathrm{E}} + U_{\mathrm{M}} + U_{\mathrm{S}} = \frac{1}{K^{\frac{2}{3}}R^{\frac{1}{3}}}[\tfrac{1}{15}P_1^{\frac{5}{3}} + \tfrac{1}{3}P_0^2P_1^{-\frac{1}{3}}] - \frac{1}{K^{\frac{2}{3}}R^{\frac{1}{3}}}\left[\frac{P_0P_1^{\frac{2}{3}}}{3} + \tfrac{2}{3}P_0^2P_1^{-\frac{1}{3}}\right] - \gamma\pi\frac{R^{\frac{2}{3}}P_1^{\frac{2}{3}}}{K^{\frac{2}{3}}}. \tag{15}$$

Equilibrium ensues when
$$\mathrm{d}U_{\mathrm{T}}/\mathrm{d}a_1 = 0.$$

This is equivalent to
$$\mathrm{d}U_{\mathrm{T}}/\mathrm{d}P_1 = 0$$
$$\frac{\mathrm{d}U_{\mathrm{T}}}{\mathrm{d}P_1} = \frac{1}{K^{\frac{2}{3}}R^{\frac{1}{3}}}[\tfrac{1}{9}P_1^{\frac{2}{3}} - \tfrac{1}{9}P_0^2P_1^{-\frac{4}{3}} - \tfrac{2}{9}P_0P_1^{-\frac{1}{3}} + \tfrac{2}{9}P_0^2P_1^{-\frac{4}{3}}] - \tfrac{2}{3}\gamma\pi\frac{R^{\frac{2}{3}}}{K^{\frac{2}{3}}}P_1^{-\frac{1}{3}}$$
$$= \frac{P_1^{-\frac{4}{3}}}{9K^{\frac{2}{3}}R^{\frac{1}{3}}}[P_1^2 - P_0^2 - 2P_1P_0 + 2P_0^2 - 6\gamma\pi RP_1]. \tag{16}$$

Therefore at equilibrium
$$\left.\begin{aligned} P_1^2 - 2P_1(P_0 + 3\gamma\pi R) + P_0^2 = 0, \\ P_1 = P_0 + 3\gamma\pi R \pm \sqrt{\{(P_0 + 3\gamma\pi R)^2 - P_0^2\}}. \end{aligned}\right\} \tag{17}$$

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By examining the second differential of the total energy it may be shown that for equilibrium to be stable it is necessary to take the positive sign so that

$$P_1 = P_0 + 3\gamma\pi R + \sqrt{\{6\gamma\pi R P_0 + (3\gamma\pi R)^2\}}. \tag{18}$$

Equation (18) shows that the apparent Hertz load P_1 acting between two elastic bodies of surface energy γ is bigger than the applied load P_0.

The Hertz equation, modified to take into account the surface energy effect is

$$a^3 = \frac{R}{K}(P + 3\gamma\pi R + \sqrt{\{6\gamma\pi R P + (3\gamma\pi R)^2\}}. \tag{19}$$

When $\gamma = 0$ this reverts to the simple Hertz equation $a^3 = RP/K$. At zero applied load the contact area is finite and given by

$$a^3 = R(6\gamma\pi R)/K. \tag{20}$$

When the applied load is made negative the contact radius decreases. For a real solution to be obtained to equation (19)

$$6\gamma\pi RP \leqslant (3\gamma\pi R)^2,$$
$$P \geqslant -\tfrac{3}{2}\gamma\pi R.$$

Separation of the spheres will just occur when

$$P = -\tfrac{3}{2}\gamma\pi R \tag{21}$$

which we note is independent of the elastic modulus.

The theoretical distribution of surface stress which has been used in this analysis, shown by curve B in figure 1*b*, calls for comment. It rises to a tension of infinite magnitude at the edge of the contact circle. In reality, of course, this stress is not achieved; the surfaces will separate slightly near the edge, thereby allowing the elastic stresses to fall to a finite value in equilibrium with the adhesive forces between the surfaces. However, from the point of view of calculating the stored elastic energy U_E, expressed in equation (12), the fact that the assumed distribution of stress differs from the true (but unknown) distribution close to the edge of contact introduces a negligible error. This approximation, and indeed the approach followed in this analysis, is similar to that used by Griffith in his criterion for the propagation of a brittle crack.

Experimental

When studying the adhesive contact between solid bodies it is important that good contact be made between the surfaces. Small interfacial gaps, due to surface asperities or dust particles, will strongly influence the adhesion because attractive surface forces decrease rapidly with increasing separation. To minimise this difficulty, experiments were carried out on optically smooth surfaces of materials with a low elastic modulus. Asperities on these were of the order of 20 nm. If the elastic

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modulus of the material is sufficiently low, the force required to flatten these asperities will be less than the surface attractive force. Therefore intimate contact ensures.

Rubber and gelatine surfaces were used for our contact studies. Roberts (1968*b*) has shown how rubber may be moulded using glass formers to produce optically smooth surfaces of low Young modulus. In the present tests it was 8×10^6 dyn cm^{-2}. Smooth elastic surfaces of even lower modulus were made by casting 25 % aqueous solutions of gelatine in glass moulds. The modulus in this case was 3×10^5 dyn cm^{-2}. Spherical surfaces of these materials were brought into contact and measurements made of the diameter of the circle of contact through a low power optical microscope. Measurements were made for both dry and lubricated surfaces.

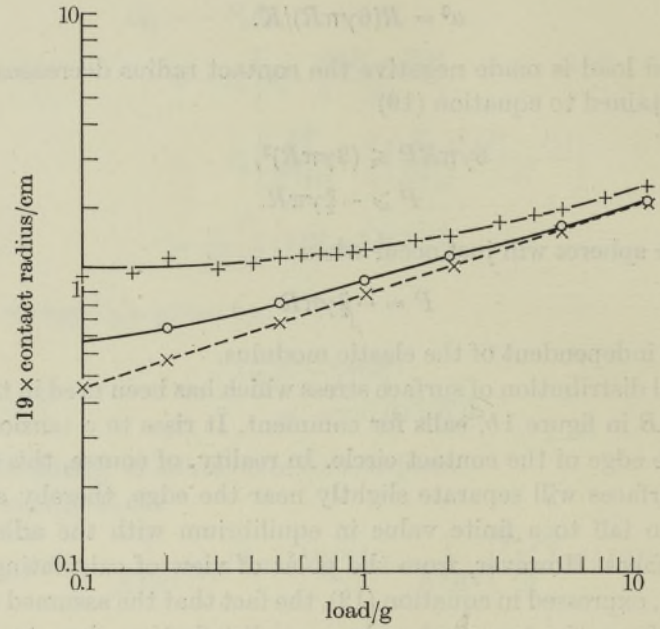


FIGURE 2. Results for two rubber spheres ($R_1 = R_2 = 2.2$ cm) in dry and lubricated contact. +, Dry contact; ○, water contact; ×, SDS solution contact; ---, Hertz theory; — modified theory.

Results for the contact between two equal rubber spheres of radius 2.2 cm are shown in figure 2. Equation (19) was found to fit the results when the dry contact total surface energy γ was taken to be 71 ± 4 erg cm^{-2}. This corresponds to an energy of 35 erg cm^{-2} for each dry rubber surface. Immersion of the surfaces in water reduced the contact size at any given load. The energy of a single rubber water interface was found (from figure 2) to be 3.4 ± 0.2 erg cm^{-2}. When the contact was immersed in a 0.01 molar solution of sodium dodecyl sulphate (SDS) the results closely agreed with the Hertz theory down to the lowest loads measured.

It should be possible to correlate these measurements of interfacial surface energy

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with surface wettability by using the Young equation (see Poynting & Thompson 1920) which may be written as

$$\gamma_{S/V} = \gamma_{S/L} + \gamma_{L/V} \cos\theta, \tag{22}$$

where $\gamma_{S/V}$ is the energy of solid surface,

$\gamma_{S/L}$ the energy of solid/liquid interface,

$\gamma_{L/V}$ the energy of liquid surface

and θ the contact angle of liquid droplet on solid surface.

The Young equation yields a value of $64 \pm 2°$ for the contact angle of a water drop resting on the rubber surface. The experimentally measured value (by direct optical observation) was $66 \pm 1°$. From figure 2 one can place an upper limit of $0.05\,\mathrm{erg\,cm^{-2}}$ on the interfacial energy between rubber and SDS solution. If this value is substituted into the Young equation $\cos\theta$ exceeds 1, indicating that a drop of SDS solution should spread over the rubber surface. This indeed is observed experimentally. As far as we are aware this is the first time that the Young equation has been verified for contact angles greater than zero. The surface energy for the contact mica-water has been determined (see Bowden & Tabor 1964) but water wets mica, hence $\cos\theta = 1$. Until now no energy values have been obtained for solid hydrocarbon surfaces which are imperfectly wetted.

The analysis presented in this paper predicts (equation (19)) that at zero applied load the contact area between surfaces should be finite and decrease as the load is made negative until a point is eventually reached at which the surfaces separate, this point being defined by equation (21). The results for small positive and negative loads applied between the surfaces of a rubber sphere (radius of curvature 2.2 cm) and a rubber flat are shown in figure 3. Agreement with theory is reasonable for loads greater than -0.3 g assuming an energy of $34\,\mathrm{erg\,cm^{-2}}$ for each rubber surface. However, near the 'pull-off' load, agreement is less satisfactory and requires comment. During the course of these experiments it became apparent that the contact equilibrium was not established instantaneously. After loading surfaces together the contact radius continued to change for some time. This may be due to viscous peeling forces which are rate dependent and have not been accounted for in our theory. At high normal loads equilibrium was reached in a few seconds but as the load approached zero this increased to minutes. At negative loads near the 'pull-off' value surface peeling was so slow that it was difficult to decide when equilibrium had been reached. In our preliminary negative load experiments the pull-off load was found to vary widely from $\frac{1}{2}$ to 5 g. However a technique was devised to assist with this problem. The experimental points shown in figure 3 were obtained in the following way and were reasonably reproducible. Surfaces were brought together very gently under zero load and after the contact circle had been established a small negative load was applied through a lever arm system together with a negative 'overload' through a micrometer screw connected directly by mounting arrangements to one of the rubber surfaces. The 'overload' could be made

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large to speed the attainment of equilibrium and then relaxed. With experience a point could be found just beyond equilibrium, the 'overload' relaxed and surfaces allowed to sink back into equilibrium under the influence of surface forces. The technique was more successful with gelatine surfaces than rubber, because for gelatine viscous forces were less dominant.

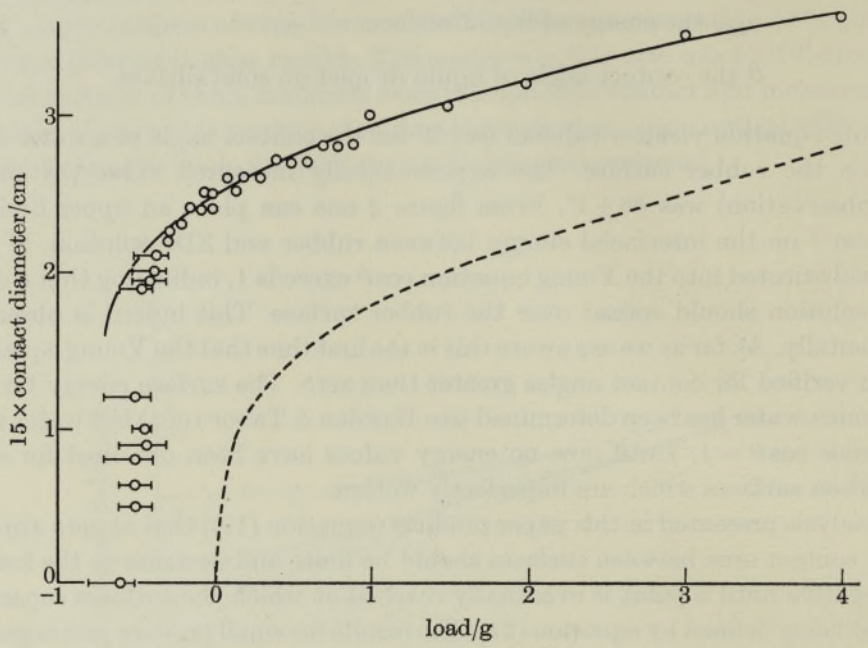


FIGURE 3. Results for rubber sphere (R = 2.2 cm) in dry contact with rubber flat under small positive and negative loads. ○, Contact results; ---, Hertz theory; —, modified theory.

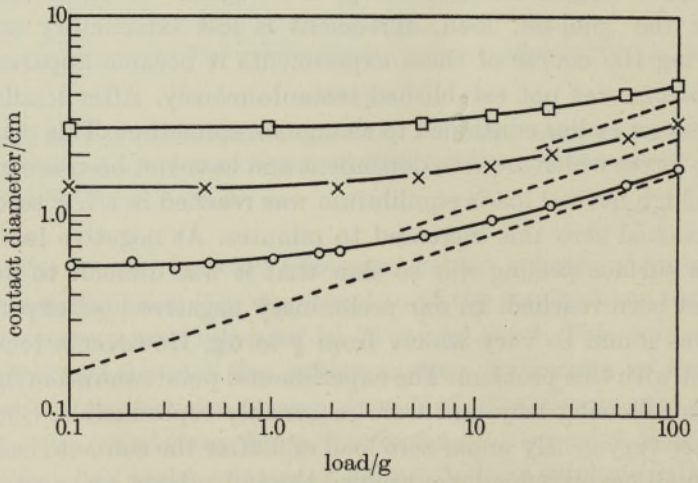


FIGURE 4. Dry contact results for gelatine spheres of different radius of curvature pressed against a Perspex flat. ○, 2.45 cm radius; ×, 7.9 cm radius; □, 25.5 cm radius; ---, Hertz theory; —, modified theory.

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Results for the contact between smooth spherical gelatine surfaces and a Perspex flat are shown in figure 4. Only dry observations were carried out but once again the measurements could be correlated with equation (19) by assuming a value of γ for the contact. Some variation in the surface energy from one gelatine moulding to another was observed, the values for gelatine on Perspex ranging from 80 to 140 erg cm^{-2} with a mean of 105 ± 10 erg cm^{-2}. The negative load measurements (see figure 5) were in close agreement with theory.

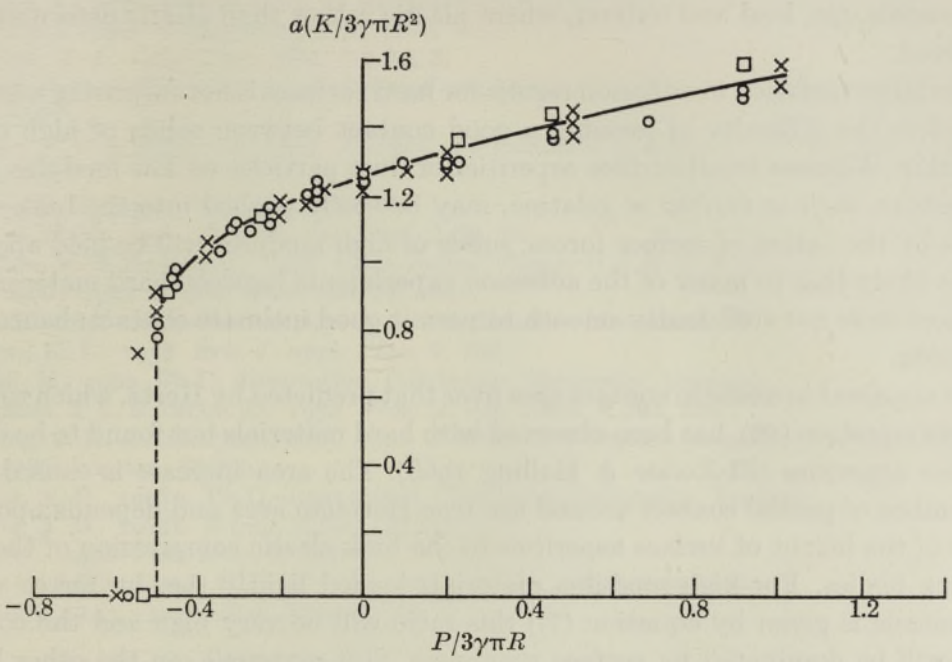


FIGURE 5. Results for gelatine spheres in dry contact with flat Perspex under small positive and negative loads plotted non-dimensionally. ○, 2.45 cm radius; ×, 7.9 cm radius; □, 25.5 cm radius; —, modified theory.

DISCUSSION

The modified Hertz theory (equation (19)) appears to account for the observed contact results between smooth surfaces of rubber and gelatine. It predicts both a contact radius at light loads greater than the calculated Hertz radius and also the negative load needed to separate bodies. In addition the effect of interposing different 'contaminating' liquids can be explained in terms of the reduction of surface energy.

In view of this it is interesting to reconsider some of the published measurements of the adhesion between solid bodies. Many studies have been reported of the adhesion between metals and glass. In some studies strong adhesion was found, in others none. For example, Tomlinson (1928) found strong adhesion between dry glass surfaces, which he attributed to molecular attraction, and Bradley (1932) found good adhesion between quartz and sodium pyroborate spheres. On the other

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hand, Budgett (1911), Stone (1930) and McFarland & Tabor (1950) found that adhesion between dry glass spheres was small, but when a thin film of water was present strong adhesive forces could be observed. This was shown to be due to the surface tension forces of a thin liquid meniscus. Kendall (1969) found reasonably strong adhesion between dry glass surfaces and was able to fit his 'greater-than-Hertz' contact radius results to equation (19). In support of their adhesion theory of friction, Bowden & Tabor (1950) have made careful measurements to find a force of adhesion between metal surfaces but little could be observed except for the very soft metals, tin, lead and indium, where plastic rather than elastic deformation is involved.

The large variation in adhesion results for hard surfaces is not surprising when one considers the difficulty of forming a good contact between solids of high elastic modulus. Whereas small surface asperities or dust particles on low modulus solids in contact, such as rubber or gelatine, may be easily pushed into the bulk of the solids by the action of surface forces, solids of high modulus will be held apart. It seems likely that in many of the adhesion experiments between hard materials the surfaces were not sufficiently smooth to permit good intimate contact; hence poor adhesion.

An apparent increase in contact area over that predicted by Hertz, which roughly follows equation (19), has been observed with hard materials but found to be due to surface asperities (El-Refaie & Halling 1968). The area increase is caused by a penumbra of partial contact around the true Hertzian area and depends upon the ratio of the height of surface asperities to the bulk elastic compression of the contacting bodies. For high modulus materials loaded lightly (i.e. by forces whose magnitude is given by equation (7)) this ratio will be very high and the contact area will be dominated by surface roughness. Soft materials, on the other hand, deform to a much greater extent under the same load so that the influence of surface roughness is negligible by comparison. Bodies made of rubber or gelatine are so soft that when pressed together, provided they are reasonably smooth, the real approaches the geometrical area of contact. Under light loads the magnitude of surface attractive forces is comparable to the force pressing such surfaces together.

Conclusion

The significance of attractive surface forces in lightly loaded contacts has been demonstrated and a modified Hertz equation produced to account for them. The new equation shows that there is a finite contact area between surfaces under zero load and it predicts the external force required to separate two bodies of given surface energy and geometry. The equation is in good agreement with observations made on spherical and flat surfaces of rubber and gelatine. However, the deformation produced by attractive surface forces between solids of high elastic modulus, such as metals and glass, is so small that its measurement is masked by the unavoidable roughness of the surfaces. With soft elastic materials this difficulty does not arise

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and the new theory provides a satisfactory explanation of the adhesion and enhanced area of contact in terms of surface energy. This approach may thus provide a method of determining the interfacial surface energy of solids if one of the bodies is highly deformable.

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